

Lecture 5 — The Delta Function and Green's Functions

18.085 Computational Science and Engineering I • MW 9:30–11, 6-120 • Wednesday, September 23, 2026

After this lecture you should be able to

- Construct $\delta(x)$ as the limit of a family of ordinary functions, and say what it means for that limit to have unit area.
- Use the sifting property to read $\delta(x - x_0)$ as an infinitely fine filter that samples a function at one point.
- Identify the step, ramp, and quadratic spline as the successive antiderivatives of δ .
- Solve a differential equation with a point load, and assemble the solution for a general load from the Green's function by superposition.

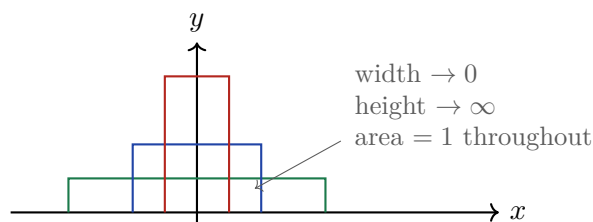
1 Physical intuition: a finite amount of stuff in no space at all

The delta function models *point concentrations*: a finite amount of “stuff” packed into an infinitesimal region of space or time. Point masses, point charges and instantaneous impulses are all of this kind. In each case the total (mass, charge, momentum transferred) is a perfectly good finite number, but the region carrying it has shrunk to nothing.

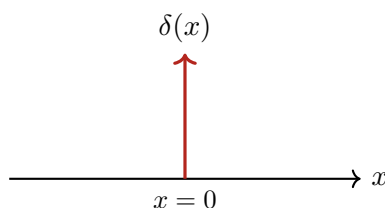
Nothing in ordinary calculus describes that. A function with a finite value at one point and zero elsewhere integrates to zero, not to one. So we build the object we want as a *limit*.

2 Mathematical construction: a limit of a family of functions

Take a family of ordinary rectangle functions, each one narrower than the last, and each one scaled in height so that the area under it stays equal to 1:



Now take the limit as the rectangle width goes to 0. The height must run off to infinity to keep the area fixed, and what is left is an infinitely concentrated *spike* at the origin, the delta function $\delta(x)$:



Since every member of the family had area 1, it is clear that the limit should too:

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

The family is not unique

Rectangles are convenient but not special. We could equally take the limit of a different family of functions, for instance increasingly narrow Gaussians, which are smooth rather than piecewise constant. Every reasonable family with unit area and shrinking support gives the same object, because what survives the limit is not the shape but the one number the shape carries.

3 The sifting property

Shift the spike to sit at x_0 instead of 0, and multiply by a function f before integrating. Away from x_0 the spike contributes nothing; at x_0 it contributes all of its unit area, weighted by the value of f there. So

$$\int_{-\infty}^{\infty} f(x) \delta(x - x_0) dx = f(x_0).$$

This is the property to remember. It says $\delta(x - x_0)$ is an infinitely fine **filter** that samples a function at the single point x_0 and reports back that one value.

Key idea

δ is defined by what it *does inside an integral*, not by its values. Every computation with δ in this course is an application of $\int f(x) \delta(x - x_0) dx = f(x_0)$ and its special case $\int \delta = 1$.

It is not a function

$\delta(x)$ is not really a “proper” function, and strictly speaking it makes rigorous sense only under an integral. Asking for the number $\delta(0)$ is not a well-posed question. In engineering and physics practice we nonetheless treat it as a real function, trusting that the mathematics can be made completely rigorous if necessary, and it can, in the theory of distributions. Just do not be surprised when a manipulation that never puts δ under an integral produces nonsense.

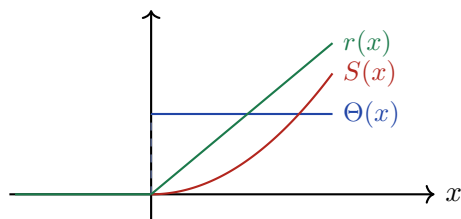
4 The antiderivatives of δ

Because δ models point concentrations, it turns up on the right-hand side of differential equations. Solving those equations means integrating δ , so it pays to know its antiderivatives. Integrating once accumulates the whole unit of area the instant we pass the origin:

$$\text{1st antiderivative: } \int_{-\infty}^x \delta(x') dx' = \Theta(x) \quad \text{the step function,}$$

and integrating again, and again,

$$\text{2nd: } \int_{-\infty}^x \Theta(x') dx' = r(x) \quad \text{the ramp,} \quad \text{3rd: } \int_{-\infty}^x r(x') dx' = S(x) \quad \text{a quadratic spline.}$$



Notice how the smoothness improves at each step: δ is not a function at all, Θ jumps, r is continuous with a corner, and S is continuously differentiable. Each integration buys one degree of regularity. That is worth remembering, because a point load in a differential equation produces a solution that is exactly this much smoother than the load.

5 Delta functions as a pointwise basis

Here is a fancy way of re-writing an arbitrary function f :

$$f(x) = \int_{-\infty}^{\infty} f(y) \delta(x - y) dy.$$

Read the right-hand side as a superposition: f is a continuous “sum” of spikes, one at every point y , with the number $f(y)$ as the coefficient. In the language of Lecture 4, the shifted deltas $\{\delta(\cdot - y)\}$ act as a basis for functions, and the coefficients in that basis are just the values of f itself.

That reformulation looks like it says nothing. It is in fact the whole method of this lecture, because we already know what linearity does to a superposition.

6 Green's functions

The setup

Suppose we have a linear differential equation. Use as the running example

$$-\frac{d^2 u}{dx^2} = f(x), \quad 0 < x < 1, \quad u(0) = 0, \quad u'(1) = 0,$$

the fixed-free problem from Lecture 2, now on the continuous side: the rod is held at $x = 0$, free at $x = 1$, and $f(x)$ is the load per unit length. Write $Lu = -u''$ for the operator. Two facts make everything below work.

1. **The operator is linear:** $L(\alpha u + \beta v) = \alpha Lu + \beta Lv$ for any constants α, β . This is Lecture 1: differentiation is a linear operator.
2. **The boundary conditions are homogeneous:** their right-hand sides are zero. So if u and v both satisfy them, so does $\alpha u + \beta v$, because $(\alpha u + \beta v)(0) = \alpha \cdot 0 + \beta \cdot 0 = 0$ and $(\alpha u + \beta v)'(1) = \alpha \cdot 0 + \beta \cdot 0 = 0$.

Together they give **superposition**: if u solves the problem with load f_1 and v solves it with load f_2 , then $\alpha u + \beta v$ solves it with load $\alpha f_1 + \beta f_2$, boundary conditions included. That is the only property of the equation used in this section.

The point-load problem

Fix a point $y \in (0, 1)$ and suppose we can solve the equation for the special case of a single unit point load at y :

$$-\frac{\partial^2 g(x, y)}{\partial x^2} = \delta(x - y), \quad g(0, y) = 0, \quad \left. \frac{\partial g}{\partial x} \right|_{x=1} = 0.$$

Note that the boundary conditions are the same ones; only the load has changed. Keep the two variables apart: x is where we look at the displacement, and y is where the load is applied. For each fixed y this is an ordinary two-point problem in x , with y as a parameter, and its solution is written $g(x, y)$. The function $g(x, y)$ is known as the **Green's function** of the problem. We find it explicitly in Section 7; here we only assume it exists and see what it buys.

Step A: chop the load into point loads

Divide $[0, 1]$ into N cells of width $\Delta y = 1/N$, with midpoints $y_1, \dots, y_N$. The load on cell k has total $f(y_k) \Delta y$, up to an error that vanishes as $\Delta y \rightarrow 0$. Replace it by a point load with the same total, sitting at the midpoint:

$$f(x) \approx f_N(x) = \sum_{k=1}^N f(y_k) \Delta y \delta(x - y_k).$$

Each cell still pushes on the rod with the same total force; only the way the force is spread inside the cell has changed.

Step B: the response to one piece

The k -th piece is the unit point load at y_k multiplied by the number $f(y_k) \Delta y$. By linearity its response is the same number times $g(x, y_k)$:

$$-\frac{\partial^2}{\partial x^2} [f(y_k) \Delta y g(x, y_k)] = f(y_k) \Delta y \delta(x - y_k),$$

and it satisfies both boundary conditions, because $g(\cdot, y_k)$ does.

Step C: add the responses

By superposition, the sum of the N responses solves the problem whose load is the sum of the N pieces:

$$u_N(x) = \sum_{k=1}^N f(y_k) g(x, y_k) \Delta y \quad \text{solves} \quad -u_N'' = f_N, \quad u_N(0) = 0, \quad u_N'(1) = 0.$$

The sum is a Riemann sum. As $N \rightarrow \infty$ it becomes an integral, and the chopped load f_N becomes the continuous superposition of spikes from Section 5, restricted to the rod:

$$f(x) = \int_0^1 f(y) \delta(x - y) dy \quad (0 < x < 1).$$

The responses add up in the same way, and the solution for a *general* right-hand side $f(x)$ is

$$u(x) = \int_0^1 f(y) g(x, y) dy$$

The load is a superposition of point loads with weights $f(y)$; the operator is linear and the boundary conditions are homogeneous; so the response is the same superposition of the point-load responses $g(x, y)$.

Step D: check the formula directly

The boxed formula can be confirmed without passing to a limit. Apply $-d^2/dx^2$ to it. The integral is over y and the derivatives are in x , so the derivatives go inside, and $f(y)$ is a constant as far as x is concerned:

$$-\frac{d^2 u}{dx^2} = \int_0^1 f(y) \left(-\frac{\partial^2 g}{\partial x^2}(x, y) \right) dy = \int_0^1 f(y) \delta(x - y) dy = f(x),$$

by the sifting property: $\delta(x - y) = \delta(y - x)$ is a spike at $y = x$. The boundary conditions go through the integral the same way:

$$u(0) = \int_0^1 f(y) g(0, y) dy = 0, \quad u'(1) = \int_0^1 f(y) \frac{\partial g}{\partial x}(1, y) dy = 0.$$

So u satisfies the equation and both boundary conditions. Here δ only ever appears under an integral, which is where the warning of Section 3 says it belongs. Once we have g explicitly, in Section 7, we repeat this check with no delta functions at all.

There is only one solution. If u and $\tilde{u}$ both solve the problem, their difference $w = u - \tilde{u}$ satisfies $-w'' = 0$, $w(0) = 0$, $w'(1) = 0$. Then $w = ax + b$, the first condition gives $b = 0$ and the second gives $a = 0$. So the formula gives *the* solution, not just a solution. For the free-free rod this argument breaks down: $w = \text{constant}$ passes both conditions $w'(0) = w'(1) = 0$, which is the continuous version of the nullspace of $\mathbf{B}$ in Lecture 2.

Why the boundary conditions must be homogeneous

Suppose the left end were held at $u(0) = a \neq 0$ and we asked each $g(\cdot, y)$ to satisfy $g(0, y) = a$. Adding up the pieces in Step C would then give $u(0) = a \int_0^1 f(y) dy$, not a . Nonzero boundary values are handled separately: solve the homogeneous-boundary problem with the Green's function, then add a solution of $-u'' = 0$ that carries the boundary values. For $u(0) = a$, $u'(1) = 0$ that extra piece is the constant a .

Solve once, solve forever

Solve the equation once, for a unit point load at an arbitrary position y , and you have solved it for every load: integrate g against f . The Green's function is the response of the system to a unit impulse, and superposition does the rest.

The discrete analogue is the inverse matrix

On a grid with spacing h , $-d^2/dx^2$ becomes a second-difference matrix $\mathbf{A}$ (with the $1/h^2$ included) and the load becomes a vector $\mathbf{f}$. A unit point load at node j is the vector $\mathbf{e}_j/h$: height $1/h$ over one cell of width h , which is the rectangle of Section 2 drawn on the grid. Its response is $\mathbf{A}^{-1}\mathbf{e}_j/h$, column j of $\mathbf{A}^{-1}$ divided by h . A general load $\mathbf{f} = \sum_j (f_j h) (\mathbf{e}_j/h)$ is a sum of point loads of strength $f_j h$, exactly as in Step A, and $\mathbf{u} = \mathbf{A}^{-1}\mathbf{f} = \sum_j (\mathbf{A}^{-1}\mathbf{e}_j/h) f_j h$ is the sum in Step C. So $\frac{1}{h}\mathbf{A}^{-1}$ is what the Green's function becomes on a grid, and $\mathbf{u} = \mathbf{A}^{-1}\mathbf{f}$

is the discrete form of $u = \int fg dy$.

7 Worked example: a point load at $x = \frac{1}{3}$

Let us actually carry this out. Solve

board

$$-\frac{d^2u}{dx^2} = \delta\left(x - \frac{1}{3}\right), \quad u(0) = 0, \quad u'(1) = 0.$$

The plan: solve the equation where it is easy (away from the load), then use four conditions to fix four constants.

Step 1: away from the load, the solution is a straight line. For $x \neq \frac{1}{3}$ the right-hand side vanishes, so $u'' = 0$ there. Integrating once gives $u' = \text{constant}$, and integrating again gives a linear function. The equation says nothing at the single point $x = \frac{1}{3}$, so the constants on the two sides need not agree:

$$u(x) = \begin{cases} Cx + D, & 0 \leq x < \frac{1}{3}, \\ Ex + F, & \frac{1}{3} < x \leq 1. \end{cases}$$

That is four unknown constants. The two boundary conditions supply two equations (Step 2) and the load supplies two more, at $x = \frac{1}{3}$ (Steps 3 and 4).

Step 2: impose the boundary conditions. The point $x = 0$ lies on the left piece, so $u(0) = C \cdot 0 + D = D$, and $u(0) = 0$ gives $D = 0$. The point $x = 1$ lies on the right piece, where $u'(x) = E$, so $u'(1) = 0$ gives $E = 0$. The solution is therefore *flat* to the right of the load:

$$u(x) = Cx \text{ on the left,} \quad u(x) = F \text{ on the right.}$$

In the language of Lecture 2, u' is the tension (the material constant is 1 here). The free end carries no tension, and nothing acts on the segment between the load and the free end, so that whole segment is unstressed.

Step 3: the load fixes the jump in slope. Integrate both sides of the equation across the load, from $\frac{1}{3} - \varepsilon$ to $\frac{1}{3} + \varepsilon$, for a small $\varepsilon > 0$. On the left, the fundamental theorem of calculus turns the integral of u'' into a difference of slopes:

$$-\int_{1/3-\varepsilon}^{1/3+\varepsilon} u'' dx = -\left[u'\left(\frac{1}{3} + \varepsilon\right) - u'\left(\frac{1}{3} - \varepsilon\right)\right] = u'\left(\frac{1}{3} - \varepsilon\right) - u'\left(\frac{1}{3} + \varepsilon\right).$$

On the right, the interval contains the spike, so $\int \delta = 1$, whatever the value of $\varepsilon > 0$:

$$\int_{1/3-\varepsilon}^{1/3+\varepsilon} \delta\left(x - \frac{1}{3}\right) dx = 1.$$

Setting the two sides equal and letting $\varepsilon \rightarrow 0$,

$$u'\left(\frac{1}{3}^-\right) - u'\left(\frac{1}{3}^+\right) = 1.$$

The slopes are C on the left and $E = 0$ on the right, so $C - 0 = 1$ and $C = 1$. The segment between the wall and the load carries the whole unit load to the wall.

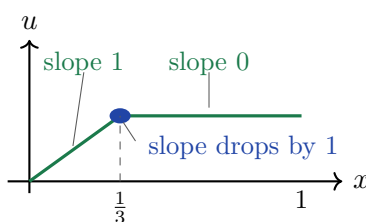
Step 4: u itself must be continuous. Suppose u jumped by an amount J at $x = \frac{1}{3}$. Then u would contain a step $J\Theta(x - \frac{1}{3})$, so u' would contain $J\delta(x - \frac{1}{3})$ and u'' would contain $J\delta'(x - \frac{1}{3})$, the derivative of a delta. The right-hand side has a δ but no δ' , so $J = 0$. (Physically, a jump in displacement would mean the rod has come apart at the load.) Matching the two pieces at $x = \frac{1}{3}$:

$$u(\frac{1}{3}^-) = C \cdot \frac{1}{3} = \frac{1}{3}, \quad u(\frac{1}{3}^+) = F, \quad \text{so} \quad F = \frac{1}{3}.$$

Therefore

$$u(x) = \begin{cases} x, & 0 \leq x \leq \frac{1}{3}, \\ \frac{1}{3}, & \frac{1}{3} \leq x \leq 1. \end{cases}$$

Check. Every requirement holds: $u(0) = 0$; u is constant on $(\frac{1}{3}, 1]$, so $u'(1) = 0$; $u'' = 0$ away from $x = \frac{1}{3}$; the slope drops from 1 to 0, by exactly 1; and both pieces equal $\frac{1}{3}$ at $x = \frac{1}{3}$.



The solution is exactly the ramp function of Section 4, bent flat at the load: it rises with slope 1 until it reaches the point source, loses all of its slope there, and coasts. Compactly, $u(x) = x - r(x - \frac{1}{3})$. This form can be checked with the antiderivatives of Section 4: $r'' = \Theta' = \delta$, so $-u'' = -(0 - \delta(x - \frac{1}{3})) = \delta(x - \frac{1}{3})$, and $u'(1) = 1 - \Theta(\frac{2}{3}) = 1 - 1 = 0$.

What a point load looks like

$-u'' = \delta(x - y)$ says: the graph of u is straight everywhere except at $x = y$, where it has a **corner**, and the size of the drop in slope across that corner is exactly the strength of the load. The delta did not survive into the answer; it survived as a kink.

The Green's function itself. Nothing above used the value $\frac{1}{3}$ in an essential way. Repeat the four steps with the load at a general point $y \in (0, 1)$:

Step 1: $g(x, y) = Cx + D$ for $0 \leq x < y$ and $g(x, y) = Ex + F$ for $y < x \leq 1$.

Step 2: $g(0, y) = D = 0$, and $\frac{\partial g}{\partial x}(1, y) = E = 0$.

Step 3: Jump in slope: $C - E = 1$, so $C = 1$.

Step 4: Continuity at $x = y$: $C \cdot y = E \cdot y + F$, so $F = y$.

Hence

$$g(x, y) = \begin{cases} x, & x \leq y, \\ y, & x \geq y, \end{cases} \quad \text{that is,} \quad g(x, y) = \min(x, y).$$

The constant F depends on where the load sits, which is why g is a function of both variables. At $y = \frac{1}{3}$ this is the answer above. So for any load f on the fixed-free bar,

$$u(x) = \int_0^1 f(y) \min(x, y) dy.$$

Checking the formula without delta functions. Split the integral at $y = x$, where $\min(x, y)$ changes from y to x :

$$u(x) = \int_0^x f(y) y \, dy + \int_x^1 f(y) x \, dy = \int_0^x y f(y) \, dy + x \int_x^1 f(y) \, dy.$$

Differentiate, using $\frac{d}{dx} \int_0^x \phi(y) \, dy = \phi(x)$, $\frac{d}{dx} \int_x^1 \phi(y) \, dy = -\phi(x)$ and the product rule:

$$u'(x) = x f(x) + \int_x^1 f(y) \, dy - x f(x) = \int_x^1 f(y) \, dy, \quad u''(x) = -f(x).$$

So $-u'' = f$, $u(0) = 0 + 0 = 0$, and $u'(1) = \int_1^1 f \, dy = 0$. Every step is ordinary calculus. The middle formula also has a physical reading: the tension $u'(x)$ at x equals the total load hanging beyond x .

Example: a uniform load. For $f \equiv 1$ (the bar under its own weight, in units where the weight per unit length and the material constant of Lecture 2 are both 1),

$$u(x) = \int_0^x y \, dy + x \int_x^1 dy = \frac{x^2}{2} + x(1 - x) = x - \frac{x^2}{2}.$$

Check: $-u'' = 1$, $u(0) = 0$, and $u'(1) = 1 - 1 = 0$. The largest displacement is at the free end, $u(1) = \frac{1}{2}$.

g is symmetric

$\min(x, y) = \min(y, x)$: the displacement at x due to a unit load at y equals the displacement at y due to a unit load at x . This is the continuous shadow of the fact that our second-difference matrices are symmetric, and their inverses with them. When we build $\mathbf{A}^T \mathbf{C} \mathbf{A}$ in Lecture 12 that symmetry will come back as a statement about energy.

Where this is going

We solved this problem by breaking the load into point sources, the basis in which the *load* is trivial, and adding up the responses. Next time we look for the other natural basis: the one in which the *operator* is trivial. Eigenvectors are the directions a matrix does not mix, and expanding in them turns matrix multiplication into multiplication by a number, exactly as expanding in deltas turned a general load into a pile of spikes.

Reading

Strang, *Computational Science and Engineering*, §1.4.